

AWS GPU Deployment Guide — Final Evolution Lab

Overview

This guide covers deploying the FEL pipeline to an AWS G5.2xlarge instance with NVIDIA A10G GPU for:

- Unreal Engine 5 compilation and cooking
- CV preprocessing with GPU acceleration
- Pixel Streaming server
- Complete build pipeline execution

Instance Specifications

Component	G5.2xlarge Spec
GPU	NVIDIA A10G (24GB VRAM)
vCPUs	8
RAM	32 GB
Storage	500 GB gp3 (configured)
Network	Up to 10 Gbps
Cost	~\$1.21/hr on-demand

Prerequisites

1. **AWS Account** with EC2 permissions
2. **AWS CLI** installed and configured (`aws configure`)
3. **GitHub account** linked to Epic Games (for UE5 source)
4. **SSH key pair** (auto-created or existing)

Quick Start

Step 1: Configure Environment

```
cp .env.template .env
# Edit .env with your AWS credentials and GitHub PAT
```

Required `.env` variables:

```
AWS_ACCESS_KEY_ID=AKIA...
AWS_SECRET_ACCESS_KEY=...
AWS_REGION=us-east-1
GITHUB_PAT=ghp_...
```

Step 2: Provision GPU Instance

```
python3 scripts/aws_deployment/provision_g5_instance.py --launch
```

This will:

- Create SSH key pair (`~/.ssh/fel-gpu-key.pem`)
- Set up security group with required ports
- Launch G5.2xlarge with Ubuntu 22.04
- Install NVIDIA drivers, CUDA, Vulkan (via user-data)
- Save instance info to `~/.fel-gpu-instance.json`

Step 3: Deploy Project to GPU

```
# Wait ~5 minutes for instance initialization
bash scripts/aws_deployment/deploy_to_gpu.sh <GPU_IP>
```

Step 4: Run Pipeline on GPU

```
# SSH into GPU instance
ssh -i ~/.ssh/fel-gpu-key.pem ubuntu@<GPU_IP>

# Navigate to project
cd /home/ubuntu/fel-workspace/rork-final-evolution-lab

# Run complete pipeline
bash scripts/ue5_setup/fel_complete_pipeline.sh
```

Step 5: Monitor Progress

From your local machine:

```
python3 scripts/aws_deployment/monitor_pipeline.py --remote <GPU_IP> --dashboard
```

Step 6: Retrieve Builds

```
# Download Linux server build
scp -i ~/.ssh/fel-gpu-key.pem -r ubuntu@<GPU_IP>:/home/ubuntu/fel-workspace/rork-
final-evolution-lab/Builds/LinuxServer/ ./Builds/

# Download iOS build files (if cooked on Linux)
scp -i ~/.ssh/fel-gpu-key.pem -r ubuntu@<GPU_IP>:/home/ubuntu/fel-workspace/rork-
final-evolution-lab/Builds/iOS/ ./Builds/
```

Step 7: Terminate Instance (IMPORTANT!)

```
# Check status
python3 scripts/aws_deployment/provision_g5_instance.py --status

# Terminate to stop charges
python3 scripts/aws_deployment/provision_g5_instance.py --terminate --instance-id i-xxxxx
```

Security Group Ports

Port	Protocol	Purpose
22	TCP	SSH
80	TCP	HTTP
443	TCP	HTTPS
3000	TCP	Frontend web app
3478	TCP/UDP	TURN/STUN (WebRTC)
8888	TCP	Pixel Streaming signalling
8889	TCP	Pixel Streaming HTTPS

Cost Optimization

Spot Instances (60-70% savings)

Modify `provision_g5_instance.py` to use spot pricing:

```
# Add to launch command
"--instance-market-options", json.dumps({
    "MarketType": "spot",
    "SpotOptions": {
        "MaxPrice": "0.50", # Set max hourly price
        "SpotInstanceType": "one-time"
    }
})
```

Estimated Pipeline Costs

Phase	Estimated Time	Cost (On-Demand)
CV Preprocessing	~30 min	~\$0.60
UE5 Build	~2-4 hours	~\$2.42-\$4.84
Cook Linux	~1-2 hours	~\$1.21-\$2.42
Cook iOS	~1-2 hours	~\$1.21-\$2.42
Total	~5-9 hours	~\$6-\$11

Troubleshooting

Instance Won't Launch

- Check service quotas: `aws service-quotas get-service-quota --service-code ec2 --quota-code L-3819A6DF`
- G5 instances may not be available in all AZs — try different subnet
- Request quota increase if needed

GPU Not Detected After Launch

- Wait for user-data script to complete (~10 minutes)
- Check setup log: `ssh ... 'cat /var/log/fel-setup.log'`
- May need reboot: `ssh ... 'sudo reboot'`

UE5 GitHub Clone Fails

- Verify `GITHUB_PAT` is set correctly
- Ensure GitHub account is linked to Epic Games at: <https://www.unrealengine.com/en-US/ue-on-github>
- Test: `curl -H "Authorization: token YOUR_PAT" https://api.github.com/repos/EpicGames/UnrealEngine`

Out of Disk Space

- Default 500GB should be sufficient
- Check: `df -h`
- Clean UE5 intermediate: `rm -rf Engine/Intermediate`

See [TROUBLESHOOTING.md](#) (TROUBLESHOOTING.md) for more details.